Rectifiers, Zener regulators, clippers

Answers and worked solutions · Day 05 · Wed 28 Oct · pack E3

5. Practice

From your course files

WS3 Q9 (course question).

(a) $V_m = \sqrt{2} \times 12 = 16.97$ V and $V_P = V_m - V_D = 16.17$ V. Half-wave, so $V_{rpp} = V_P/(fCR_L)$:

$$C = \frac{V_P}{V_{rpp} f R_L} = \frac{16.17}{1 \times 50 \times 100} = \mathbf{3.23\ mF}$$

(b) $V_{Odc} = V_P - \frac{1}{2}V_{rpp} = 16.17 - 0.5 =$ **15.67 V**. $I_L = 15.67/100 =$ **156.7 mA**. PIV $\approx 2V_m =$ **33.94 V** (official; see the note above for 33.14 V).

WS3 Q10 (course question).

Zener model: $V_O = V_{zk} + I_Z r_z$. KCL at the output node, $I_s = I_Z + I_L$:

$$\frac{15 - V_O}{100} = \frac{V_O - 10}{20} + \frac{V_O}{500} \;\Rightarrow\; 6.2V_O = 65 \;\Rightarrow\; V_O = \mathbf{10.48\ V}$$

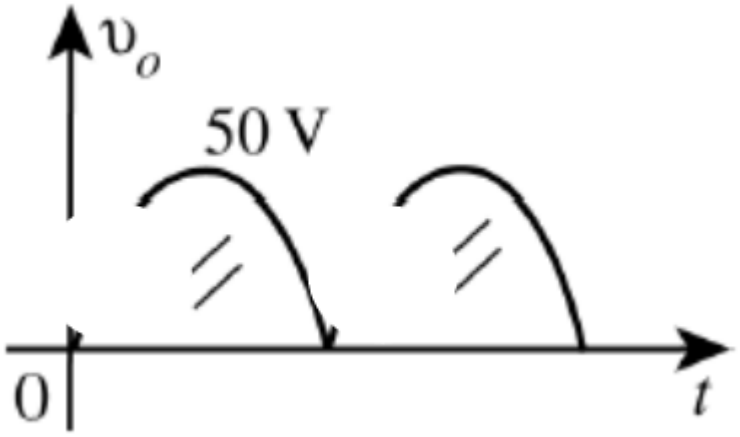
$I_s = (15 - 10.48)/100 =$ **45.2 mA**, $I_L = 10.48/500 =$ **20.97 mA**, $I_Z = (10.48 - 10)/20 =$ **24.2 mA**, $P_Z = V_O I_Z =$ **254 mW** (official 251.5 mW after rounding I_Z to 24 mA).

WS3 Q11 (course question).

Ideal diodes. In each half-cycle one diode conducts, and the circuit reduces to one 2.2 kΩ bridge resistor in series with the 2.2 kΩ load, so it is a 2.2/2.2 divider:

$$V_{o,peak} = \frac{2.2}{2.2 + 2.2} \times 100 = \mathbf{50\ V}$$

The load current has the same direction in both half-cycles, so v_o is a full-wave rectified sine of peak 50 V.



WS3 Q11 official output sketch

$$V_{dc} = 0.636 \times 50 = \mathbf{31.8\ V}$$

WS3 Q12 (course question).

Worked in full in Worked example 4 above (this is also practice test Q5). $v_i = +20$ V: diode reverse biased, $v_o = 0$ **V**. $v_i = -5$ V: diode on, $+5 - 4 + v_o = 0$, so $v_o = -1$ **V**. v_o is a square wave between 0 and −1 V.

WS3 Q13 (course question).

Worked in full in Worked example 3 above (this is also practice test Q4). $V_Z = 12$ **V.** At full load $R_L = 12/0.2 = 60\ \Omega$ and $I_Z \to 0$, so $\dfrac{60 \times 16}{60 + R_S} = 12$ gives $R_S = 20\ \Omega$. At no load the Zener takes 200 mA, so it needs $P_Z \geq 2.4$ W.

WS3 Q14 (course question).

Worked in full in Worked example 1 above. $C = V_P/(V_r f R) =$ **83.3 µF**.
$\omega \Delta t = \sqrt{2V_r/V_P} = 0.2$ rad, so the diode conducts for **3.18 %** of the cycle.
$i_{Dav} = I_L(1 + \pi\sqrt{2V_P/V_r}) =$ **324 mA** and $i_{Dmax} = I_L(1 + 2\pi\sqrt{2V_P/V_r}) =$ **638 mA**.

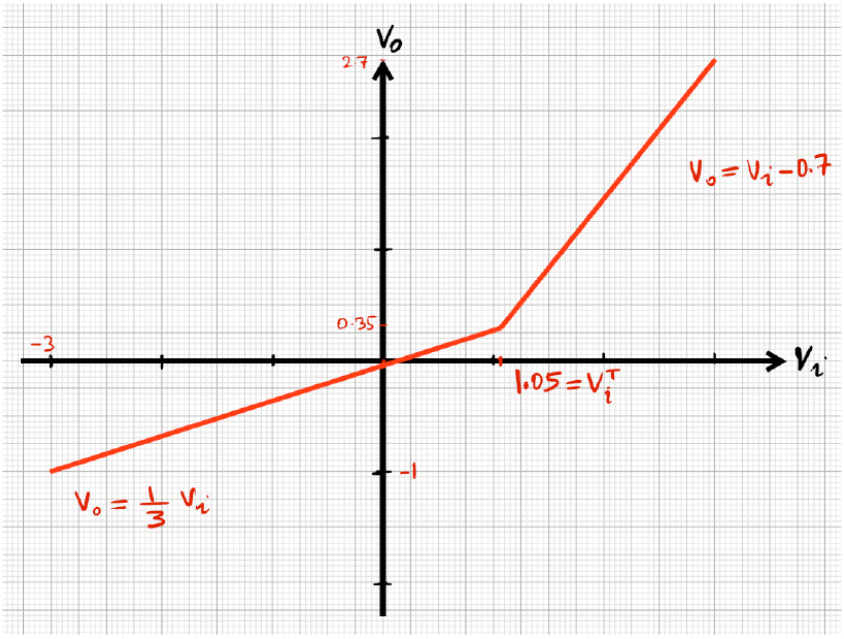
WS3 Q18 (course question).

The diode sits across R_1, so while it is off $v_d = v_{R1} = V_i \dfrac{R_1}{R_1 + R_2} = \dfrac{V_i}{1.5}$. It turns on when this reaches 0.7 V:

$$\frac{v_{iT}}{1.5} = 0.7 \ \Rightarrow\ v_{iT} = \mathbf{1.05\ V}$$

- $V_i < 1.05$ V (diode off): divider, $V_o = V_i \dfrac{0.5}{1.5} = \dfrac{V_i}{3}$.
- $V_i > 1.05$ V (diode on, 0.7 V source): KVL, $V_o = V_i - 0.7$.

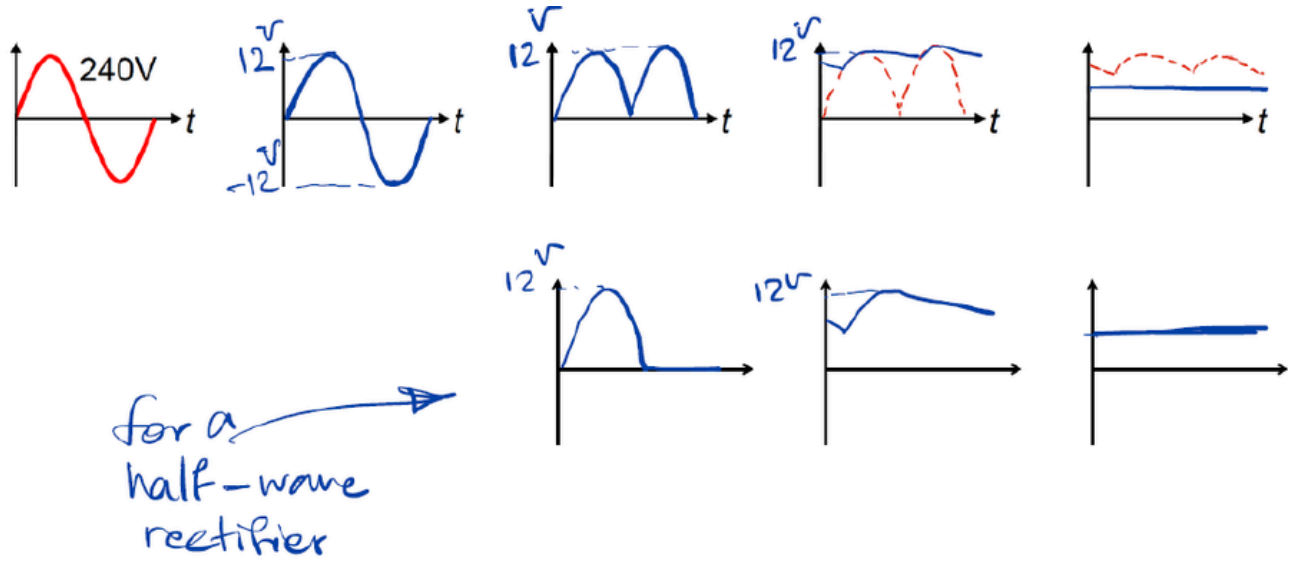
Endpoints: $V_o = -1.67$ V at $V_i = -5$ V, $V_o = 0.35$ V at the break, $V_o = 4.3$ V at $V_i = +5$ V.



WS3 Q18 official plot (drawn over a narrower range than the question asks; extend both lines to ±5 V)

WS3 Q19 (course question).

Sketches only. 240 V ac sine → transformer: 12 V peak sine → rectifier: pulsating dc (bridge: a hump every half-cycle, at $2f$; half-wave: one hump per cycle, at f) → capacitor: dc with a sawtooth ripple (the half-wave ripple is larger because it refills only once per cycle) → regulator: flat 5 V dc.



WS3 Q19 official sketches: top row bridge, bottom row half-wave

WS3 Q20 (course question).

Half-wave: current flows only for $0 < \omega t < \pi$.

1. $I_m = \dfrac{V_m}{r_f + R} = \dfrac{50}{820} =$ **61.0 mA**
2. $I_{dc} = \dfrac{1}{2\pi}\displaystyle\int_0^{\pi} I_m \sin\omega t \, d(\omega t) = \dfrac{I_m}{\pi} =$ **19.4 mA**
3. $V_{m,L} = 50 \times \dfrac{800}{820} =$ **48.8 V**
4. $V_{DC,L} = RI_{dc} =$ **15.5 V**
5. $I_{rms} = \sqrt{\dfrac{I_m^2}{2\pi}\displaystyle\int_0^{\pi} \sin^2 \omega t \, d(\omega t)} = \dfrac{I_m}{2} =$ **30.5 mA**
6. $V_{L,rms} = V_{m,L}/2 =$ **24.4 V**
7. The diode sees $r_f I_m \sin\omega t$ forward and $V_m \sin\omega t$ reverse, so $V_{D,DC} = \dfrac{1}{2\pi}\left[\displaystyle\int_0^{\pi} r_f I_m \sin\omega t \, d(\omega t) + \int_{\pi}^{2\pi} V_m \sin\omega t \, d(\omega t)\right] = \dfrac{r_f I_m - V_m}{\pi} = -$ **−15.5 V** (equal and opposite to the load's dc voltage, as KVL requires).
8. $P_{AC,i} = I_{rms}^2 (r_f + R) =$ **0.762 W**
9. $P_{DC,o} = I_{dc}^2 R =$ **0.301 W**
10. $\eta = P_{DC,o}/P_{AC,i} =$ **39.5 %**

WS3 Q21 (course question).

Two diodes conduct in series at any moment, so the path resistance is $2r_f + R$.

1. $I_m = \dfrac{240}{2(20) + 460} =$ **0.48 A**
2. $I_{dc} = \dfrac{2I_m}{\pi} =$ **0.306 A**

3. $V_{m,L} = 240 \times \dfrac{460}{500} =$ **220.8 V**
4. $V_{DC,L} = RI_{dc} =$ **140.6 V**
5. PIV $= V_m =$ **240 V**
6. Two output pulses per input cycle: $\omega_o = 2 \times 314 = 628$ rad/s, so $f_o = 628/2\pi =$ **99.95 Hz**
7. Each diode conducts half the cycle, so its $I_{rms} = I_m/2 = 0.24$ A and $P_D = I_{rms}^2 r_f =$ **1.15 W**

WS3 Q22 (course question).

$I_R = (V_s - V_Z)/R_s$: $I_{R,max} = (11 - 4.8)/1 = 6.2$ mA and $I_{R,min} = (9 - 4.8)/1 = 4.2$ mA. KCL: $I_L = I_R - I_Z$.

- **Smallest load current.** Even with no load the Zener takes at most $I_{R,max} = 6.2$ mA, which is below $I_{Z,max} = 7$ mA. So $I_{L,min} = 0$ is allowed and $R_{L,max} = \infty$.
- **Largest load current.** It must work at the lowest supply with the Zener still at $I_{Z,min}$: $I_{L,max} = I_{R,min} - I_{Z,min} = 4.2 - 0.2 = 4$ mA. So $R_{L,min} = 4.8/4\,\text{mA} = 1.2$ **kΩ**.

New questions

N1 (easy) [4 marks].

a. $V_m = 15\sqrt{2} = 21.21$ V. Two diodes are in the path, so $V_P = 21.21 - 1.4 = \mathbf{19.81\ V}$.

b. Full-wave, so use $2f$:

$$V_r = \frac{V_P}{2fCR} = \frac{19.81}{2 \times 50 \times 1000 \times 10^{-6} \times 470} = \mathbf{0.422\ V}$$

The ripple frequency is $2f = \mathbf{100\ Hz}$.

c. $V_{O(dc)} = V_P - V_r/2 = 19.81 - 0.21 = \mathbf{19.60\ V}$. Bridge PIV $= V_S - V_D = 21.21 - 0.7 = \mathbf{20.5\ V}$, so choose a diode rated for at least about 31 V.

(If you used f instead of $2f$, you got 0.843 V. That is the trap.)

N2 (medium) [6 marks].

a. $V_m = 12\sqrt{2} = 16.97$ V and $V_P = 16.97 - 0.7 = 16.27$ V. Full-wave with a known load current:

$$C = \frac{I_L}{2fV_r} = \frac{0.5}{2 \times 50 \times 1} = \mathbf{5000\ \mu F}\ \text{(choose the next standard value up)}$$

b. $V_{O(dc)} = 16.27 - 0.5 = \mathbf{15.77\ V}$. Centre-tap PIV $= 2V_S - V_D = 2(16.97) - 0.7 = \mathbf{33.2\ V}$.

c. $\omega\Delta t = \sqrt{2V_r/V_P} = \sqrt{2/16.27} = \mathbf{0.351\ rad}$ (20.1°), so $\Delta t = 0.351/(2\pi \cdot 50) = 1.12$ ms each half-cycle. For full-wave, each diode only has to replace the charge lost in $T/2$:

$$i_{Dmax} = I_L\left(1 + 2\pi\sqrt{\frac{V_P}{2V_r}}\right) = 0.5\left(1 + 2\pi\sqrt{8.135}\right) = \mathbf{9.46\ A}$$

($i_{Dav} = 0.5(1 + \pi\sqrt{8.135}) = 4.98$ A.) Using the half-wave form from the formula sheet would give $i_{Dav} = 9.46$ A, which is twice too large. If you use it in a test, say so.

N3 (medium) [6 marks].

a. KCL at the output node:

$$\frac{V_O - 6}{5} + \frac{V_O}{1000} = \frac{12 - V_O}{150}$$

This gives $\mathbf{V_O = 6.164\ V}$. Then $I_Z = (6.164 - 6)/5 = \mathbf{32.7\ mA}$, $I_L = 6.16$ mA, $I_S = (12 - 6.164)/150 = 38.9$ mA (check: 32.7 + 6.2 = 38.9 ✓).
$P_Z = 6.164 \times 32.7\ \text{mA} = \mathbf{202\ mW}$.

b. Repeat with 13 V: $V_O = 6.196$ V, so $\Delta V_O = 32.1$ mV for $\Delta V_s = 1$ V. **Line regulation = 32.1 mV/V.** Check with the formula: $\dfrac{r_z \parallel R_L}{R_S + r_z \parallel R_L} = \dfrac{4.975}{154.975} = 0.0321$ ✓.

c. No load: $\dfrac{V_O - 6}{5} = \dfrac{12 - V_O}{150}$, which gives $V_{NL} = 6.194$ V. Then

$$\%V.R. = \frac{6.194 - 6.164}{6.164} \times 100 = \mathbf{0.48\%}$$

Compare this with the 33 % of your unregulated Prac 1 supply.

N4 (hard, chained) [10 marks].

a. $V_m = 18\sqrt{2} = 25.46$ V. $V_P = 25.46 - 1.4 = \mathbf{24.06\ V}$. The valley is $V_P - V_r = \mathbf{22.06\ V}$.

b. The worst case is the valley with full load:

$$R_S \le \frac{22.06 - 9.1}{0.050 + 0.005} = \mathbf{235.6\ \Omega}$$

Choose **220 Ω**, the next standard value below. Check: at the valley, $(22.06 - 9.1)/220 - 5\ \text{mA} = 53.9$ mA ≥ 50 mA ✓.

c. While the Zener is in breakdown, $V_O = 9.1$ V is fixed, so $I_S = (v_C - 9.1)/R_S$ depends only on the capacitor voltage. The load only changes how I_S splits between I_Z and I_L. This is the same thing your Prac 1 showed. With $V_{C,avg} = V_P - V_r/2 = 23.06$ V:

$$I_S = \frac{23.06 - 9.1}{220} = 63.4\ \text{mA}$$

The capacitor discharges into this current, and it is full-wave:

$$C = \frac{I_S}{2fV_r} = \frac{0.0634}{2 \times 50 \times 2} = 317\ \mu\text{F} \Rightarrow \mathbf{choose\ 330\ \mu F}$$

That gives $V_r = 1.92$ V ✓.

d. The worst case for the Zener is no load, where it takes all of I_S. At the average capacitor voltage, $P_Z = 9.1 \times 63.4\ \text{mA} = 0.577$ W. At the ripple peak, $I_S = (24.06 - 9.1)/220 = 68.0$ mA, so $P_{Z,max} = \mathbf{0.62\ W}$. Specify a 1 W Zener. Bridge PIV $= V_m - V_D = \mathbf{24.8\ V}$, so choose diodes rated at least 37 V (for example, 50 V parts).

e. With $V_r = 1.92$ V, using the full-wave form:

$$i_{Dmax} \approx I_S \left(1 + 2\pi\sqrt{\frac{V_P}{2V_r}}\right) = 0.0634\left(1 + 2\pi\sqrt{6.26}\right) \approx \mathbf{1.06\ A}$$

N5 (medium) [5 marks].

a. With both diodes off, no current flows in the 1 kΩ, so $v_o = v_i$.

- D_1 turns on when $v_o = 3 + 0.7 = 3.7$ V, so $v_{iT1} = +3.7$ V. Above it, $\mathbf{v_o = 3.7\ V}$ (clipped).
- D_2 turns on when $v_o = -2 - 0.7 = -2.7$ V, so $v_{iT2} = -2.7$ V. Below it, $\mathbf{v_o = -2.7\ V}$.
- For $-2.7 < v_i < 3.7$ V, $\mathbf{v_o = v_i}$.

Input and output of the double clipper in N5.

b. v_o is the sine with its top flattened at +3.7 V and its bottom flattened at −2.7 V. The current in the 1 kΩ at the positive peak is $(10 - 3.7)/1\text{k} = 6.3$ mA. At the negative peak it is $(10 - 2.7)/1\text{k} = 7.3$ mA in the other direction. The largest current is therefore **7.3 mA**, at the negative peak.

N6 (easy) [4 marks].

a. When the input is −10 V, the diode conducts and C charges to 10 V. Thereafter the output is $v_i + 10$, so it switches between $\mathbf{0\ V}$ and $\mathbf{+20\ V}$. The shape is unchanged and it is still 20 V peak-to-peak.

b. $V_m = 12\sqrt{2} = 16.97$ V. The ideal output is $2V_m = \mathbf{33.9\ V}$. With two 0.7 V drops, it is $2(16.97 - 0.7) = \mathbf{32.5\ V}$. Each diode must block $2V_m = \mathbf{33.9\ V}$.

6. Quick recall check

Question

1. Bridge : half-wave = 1 : 2. The bridge refills every $T/2$, so $V_r = V_P/(2fCR)$.
2. $C = I_L/(fV_r)$, or $V_P/(fRV_r)$.
3. Size R_S at maximum load and minimum supply: $R_S = (V_{s,min} - V_Z)/(I_{L,max} + I_{Z,min})$. Check P_Z at no load and maximum supply.
4. $2V_S - V_D$.
5. It decreases, because a wider depletion layer means a larger plate separation.